CPS 643 Final Project

Scripts

Strike

500966762 - Tyler Quach

1. Batting Scripts

These scripts have to relate to the player and their role in batting the ball to score points. The reason why I decided to use scripts to handle bat and ball collision was that it added some consistency/leeway to the batter ultimately making the game easier to pick up and more arcadey rather than simulation.

- Assets
    - Bat (model from turbo squid)
        - 4 collider capsules
            - 4 follower capsules
        - Grip

Script a. and b. Were referenced and adjusted from this tutorial

a. Bat Capsule.cs

This script is tied to 4 different capsule game objects which represent the hitbox of the bat, it does not have any collider and is solely there to stick to the bat.

b. Bat Capsule Follower.cs

This script is the main physics component of the bat as it uses rigid bodies that follow the bat to calculate the distance between the original capsule and the follower. Each bat capsule has a follower which in turn will add a sweet spot to the bat as the tip of the bat would travel the most distance in a swing as it has a longer swing circumference compared to the bottom of the bat.

c. Contact.cs

This script is tied to the bat capsule follower and grabs the velocity from the rigid body to transfer to the velocity of the ball.

d. Grab.cs

This script is tied to the bat grip (sphere collider) and is used to determine if the player is in a ready position for batting and will add another gameplay aspect of keeping your hands inline for an accurate strike. The bat will tilt from an interpolation between the position of the batting hand and off hand controllers.

e. Handedness.cs

This script is part of the controller game objects and mainly deals with managing the controllers offhand and main hand. When the game commences the bat will spawn on the main hand and the offhand will have the grab script enabled. Also when the game commences this script handles teleportation to the correct batting box.

f. Runner

This script is tied to the Runner gameObject which is a physical representation of runners in a baseball game. This prefab stores the current base that the runner is at and will update the location based on any changes to the base location

g. RunnerState

This script handles all the runners and the ability to change the state of the current runners in the game. This script is tied to the baseball diamond game object. It gets the ball distance after determining if the ball is valid and will update all the runners if the distance is a single,double, triple. And will also update the jumbotron message board and run score.

2. Audio Script

These scripts are tied to audio sources in the stadium

a. CrowdAudio.cs

Handles playing and muting crowd audio

b. InningAudio.cs

Handles playing and muting a random inning audio, when the game commences or when a batter goes out

3. Game Arena

These scripts are tied to the foul, ball, and strike zones that detect and manage the game state

- Assets
    - Left Foul Line
        - Left Foul Area
    - Right Foul Line
        - Right Foul Area
    - Ball Zone
    - Strike Zone
        - Strike Indicator Prefab

a. ballZone.cs

This script is tied to the Ball Zone game object and consist of a box collider. When the ball is thrown from the pitcher and collides with the Ball Zone the script will determine if the player has swung the bat or not, based on this it will determine and update the ball counter or strike counter. It does this by grabbing the velocity from the Bat Capsule Follower and determines if the bat has moved fast enough to be considered a swing.

b. foulZone.cs

This script is tied to the Left and Right Foul Area and consist of a box collider. When the ball is hit into the zone the script will detect if the ball passes into the foul zone and proceed the game logic as a foul ball.

c. strikeZone.cs

This script is tied to the strike zone behind the player and consist of box collider. When the pitcher pitches the ball and it collides with the box collider it will spawn in a prefab indicator on the strike zone and increase the strike count on the game state

d. fade.cs

Runs a coroutine which will remove the strike indicator after a duration of time

4. Jumbotron

These scripts handle the game state and the visuals of the

- Asset
    - Jumbotron
        - Screen (displays ball camera)
        - TV (decorative game objects)
        - Label
        - Ball Indicator
        - Strike Indicator
        - Out Indicator
        - Text
        - Runs

a. GameScore.cs

This script handles all the scores and data for the game state and has functions to reset and increase each score (Ball, Strikes, Outs) and will run functions based on if there are 3 outs or 3 strikes (turns into an out).

b. ParentIndicator.cs

This script gathers the children's cube game objects and loops through a list to manage which child should be active or not

c. ScoreActive.cs

This script is tied to each indicator child which is just a cube game object. If the cube is active then the material will change to visually signify the game state.

5. Pitcher

- Assets
    - Ball (model from turbo squid)
    - Pitching Cannon
    a. Ball.cs

This script is tied to a ball game object which is a prefab. It handles the ball state (hit, stop, destroy) for when a game event happens. Depending on these values effects game decisions that affect the game loop like when a ball made contact with the player bat or not will affect the increasing scores in the game state. Also the rigidbody in this prefab helps determine the distance velocity and physic based metrics.

b. Pitching Cannon.cs

This script is the main script for the pitcher and ties alot of the ends in the game. First the pitcher has an indicator to the player to better understand the game events occurring when batting. Then the pitching cannon will shoot the ball if the player is in a ready stance and after a random determined throw timer has passed. Once this ball has been spawned in it will have a randomly decided impulse force which would affect the trajectory of the ball. Once the ball is thrown, depending if the ball has made contact with the ball the script will handle the positioning of the camera based on the ball and will update the texture on the jumbotron to give more accessibility to the player. And while the ball is traveling from the pitch or bat the script will update the jumbotron text accordingly.

6. Selector/UI Canvas

Inspired from this tutorial

- Assets
    - Pointer
    - Canvas
    a. Selector.cs

This script is connected to the controller game object and enables the player to interact with the game menu. It grabs information from the line renderer and the custom input module to process commands.

b. VR_Input_Module.cs

This script uses cameras to detect the pointer cursor hovering over buttons and reads input from the controllers to interact with the menu ui. Using unity event systems we can use this script to trigger events that manipulate objects and game state.

c. Panel.cs

This script manages each UI panel state and connects the panel to parent panel groups. Runs functions such as show and hide.

d. MenuManager.cs

This script holds all panels in a list and manages the state of which panel and page should be shown on the main canvas menu.

e. ActiveHands.cs

This script toggles the variables and states regarding the player's preferred hand. Functionality such as selector and primary batting hand is set and determined in this script.